25900803 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCECT100 - HOME APPLIANCES AND TROUBLESHOOTING

(2024 ADMISION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **35** Questions

Each Question carries **1** mark

1. Why is thermal imaging used in equipment inspection? [U] / [CO1]

   a). To check for voltage fluctuations b). To detect overheating components

   c). To verify grounding systems d). To measure insulation resistance

2. Which material typically has the highest resistance? [K] / [CO1]

   a). Copper b). Aluminum

   c). Silver d). Rubber

3. In a simple circuit, if the voltage is increased while the resistance remains constant, what happens to the current? [U] / [CO1]

a). It decreases b). It remains the same

c). It increases d). It becomes zero

4. What is electric current? [K] / [CO1]

a). The flow of electrons b). The flow of protons

c). The flow of neutrons d). The flow of heat

5. In a DC system, which color is often used for the positive wire? [K] / [CO1]

a). Blue b). Red

c). Black d). Yellow

6. What type of cable accessory provides protection from mechanical stress? [K] / [CO1]

a). Armored sheath b). Cable tie

c). Junction box d). Connector

7. What should you do if a person is unresponsive but breathing? [U] / [CO1]

a). Place them in a standing position b). Place them in the recovery position

c). Shake them to wake them up d). Leave them alone

8. What is one common application of three-phase power systems? [U] / [CO1]

a). Household lighting b). Residential heating

c). Large industrial motors d). Small electronic devices

9. Why is it important to consider the length of the wire when selecting gauge? [U] / [CO1]

a). Affects color b). Affects voltage drop

c). Affects insulation d). No effect

10. Which material is often used for cable ties in cable management? [K] / [CO1]

a). Copper b). Nylon

c). Aluminum d). PVC

11. What happens to a fuse when excessive current flows through it? [K] / [CO1]

a). It sparks b). It blows

c). It overheats
d). It short-circuits

12. What is the primary function of an RCCB? [K] / [CO1]

a). Detect short circuits
b). Detect earth leakage
c). Regulate voltage
d). Reduce current flow

13. What is the unit of electric power? [K] / [CO1]

a). Volt
b). Ohm
c). Ampere
d). Watt

14. Which of the following tests is performed to check the effectiveness of an earthing system? [K] / [CO1]

a). Voltage drop test
b). Insulation resistance test
c). Earth resistance test
d). Continuity test

15. Which of the following devices is designed to protect against overload? [K] / [CO1]

a). Capacitor
b). Resistor
c). Circuit breaker
d). Transistor

16. What is the main symptom of a phase-to-phase fault? [U] / [CO1]

a). Overheating of equipment
b). Flickering lights
c). Voltage imbalance between phases
d). Complete power failure

17. What is the primary function of an ELCB? [K] / [CO1]

a). To detect short circuits
b). To prevent current leakage to the earth
c). To reduce the voltage
d). To balance the electrical load

18. Why is grounding important in electrical systems? [U] / [CO1]

a). It provides an alternate path for electricity
b). It increases the efficiency of the system
c). It reduces energy consumption
d). It prevents voltage drop

19. How can you check if someone is responsive? [U] / [CO1]

a). Shake them gently and shout b). Check their pulse

c). Look for breathing d). All of the above

20. Why is it dangerous to use appliances with damaged cords? [U] / [CO1]

a). They consume more electricity b). They can cause electrical shock and fires

c). They function poorly d). They reduce power consumption

21. What is the risk of using a multimeter incorrectly? [U] / [CO2]

a). Damage to the device b). Safety hazards

c). Inaccurate readings d). All of the above

22. What is the primary function of a chisel? [U] / [CO2]

a). To shape and carve wood, stone, or metal b). To cut metal

c). To drive nails into wood d). To measure precise dimensions

23. Which measurement mode is best for testing capacitors? [U] / [CO2]

a). Resistance mode b). Voltage mode

c). Capacitance mode d). Current mode

24. What is the primary function of a hammer? [U] / [CO2]

a). To cut metal b). To measure distances

c). To tighten screws d). To drive nails into materials

25. How can a faulty transformer be diagnosed? [U] / [CO2]

a). Measure voltage output b). Measure resistance

c). Both A and B d). None of the above

26. How is a diode tested with a multimeter? [K] / [CO2]

a). Measure capacitance b). Measure voltage

c). Use diode mode d). Measure resistance

27. What does a low resistance value between the drain and source of a MOSFET in both directions indicate? [U] / [CO2]

a). The MOSFET is functioning properly
b). The MOSFET is shorted
c). The MOSFET is in the ON state
d). The MOSFET is in the OFF state

28. What is the primary function of a scriber? [U] / [CO2]

a). To cut metal surfaces
b). To polish surfaces
c). To measure angles in metalworking
d). To make precise markings on materials

29. How does a digital voltmeter show readings? [U] / [CO2]

a). Analog scale
b). Digital display
c). Both
d). None of the above

30. How can you check if a potentiometer is functioning correctly? [U] / [CO2]

a). Measure resistance
b). Check for shorts
c). Both A and B
d). None of the above

31. What unit does an ammeter display current in? [K] / [CO2]

a). Volts
b). Ohms
c). Amperes
d). Watts

32. What should you do before using a clamp meter? [K] / [CO2]

a). Check battery level
b). Calibrate it
c). Ensure proper connection
d). All of the above

33. What is the primary function of a wrench? [U] / [CO2]

a). To measure the size of objects
b). To hold materials securely
c). To tighten or loosen bolts and nuts
d). To cut through metal

34. What is the key characteristic of a hacksaw blade? [U] / [CO2]

a). It is circular in shape
b). It is designed with fine teeth for cutting
c). It is used for sanding
d). It is made of wood

35. What does a multimeter check when it is in continuity mode? [U] / [CO2]

a). Voltage
b). Current
c). Resistance
d). All of the above

36. What type of screwdriver has a flat, straight blade that fits into a straight screw slot? [K] / [CO2]

a). Phillips screwdriver
b). Flathead screwdriver
c). Hex screwdriver
d). Torx screwdriver

37. What is the primary use of tweezers? [U] / [CO2]

a). To cut small materials
b). To grip and manipulate small objects
c). To hammer nails
d). To measure dimensions

38. Which part of the wire cutter is specifically designed for cutting wire? [U] / [CO2]

a). The handles
b). The pivot point
c). The cutting jaws
d). The spring mechanism

39. For twisting, pulling, and gripping small jobs, which tool is typically used? [U] / [CO2]

a). Pliers
b). Chisel
c). File
d). Spanner

40. Which shape characterizes an Allen key? [K] / [CO2]

a). Square
b). Hexagonal
c). Round
d). Flat

[35X1=35]

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